



Deep Learning & Computer Vision:

Lab session 2/2

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Synopsis

- In this lab session, exercise 4, you will train a CNN to classify skin moles into seven categories.
- You will use pytorch [1] library. You will need to turn on the GPU-accelerator HW running in cloud provided along with gmail account. Your code and data will be stored at your GDrive.
- The data have been publicly released. I have preprocessed the data for you. The data will download automatically when you first execute the `exercise_4_skin-cancer-assignment.ipynb` notebook.

Literature :

- Esteva, Andre, et al., **Dermatologist-level classification of skin cancer with deep neural networks**, Nature, vol. 542, no. 7639, p. 115, 2017 [paper] (prefer the OPEN-ACCESS PDF link)

1. a high-level library for neural network programming

1. Open a navigator and connect to your Gmail account
2. Copy the files `exercise_4_skin-cancer-assignment.ipynb` and `exercise_functions.py` from moodle Lab sessions 6 (resources) into the directory "Colab Notebooks" in your drive.
3. Open the `exercise_4_skin-cancer-assignment.ipynb` in Colab. Turn on the GPU accelerator and proceed.
4. A written report is due by the date given in your moodle.

1. This file is a python notebook `.ipynb`. Notebooks are *active* files allowing to mix images, text and code. When the code is executed the results can either be saved or displayed in the notebook.
2. Have a look on the file. Different portions of the document are structured in *cells*. You can modify a text cell by double clicking on it. You can run a code cell by clicking inside the brackets `[ ]` on top left of the cell. Once a cell has been executed it is assigned a number in the brackets. The cells should be executed in the order of appearance.
3. The output appears below each cell (or an error flag  and a message below and if there was an error.
4. Various other commands related to the code execution are available in the menu.
5. Go on to execute one by one the cells of the notebook. Do not "execute all" because your interaction is needed from time to time.

- Finish executing the script. What accuracy could you obtain ?
- Try to improve it : add more hidden layers, make the data augmentation more agressive (increase rotation, zoom, shift, add flips)
- Could you improve the accuracy ?
- A written report from this lab is due by the date indicated in the moodle.